

David Shabo

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Education

University of Windsor

Windsor, ON

Bachelor of Computer Science, Artificial Intelligence Specialization

4th Year

Work Experience

Data Solutions Group

Chesterfield, MO (Remote)

Software Developer – Intern

Sep 2025 – Dec 2025

- Worked on an internal CRM used by sales, improving how contract and renewal information was tracked and displayed.
- Cleaned up and standardized contract data to ensure renewal dates and service details were consistent.
- Wrote backend queries and logic to surface customer services approaching renewal through existing APIs.
- Updated frontend views to display upcoming renewals in simple tables and filtered dashboards.
- Tested CRM changes with sample and real data to validate renewal timelines and edge cases.
- Coordinated with engineers and systems staff to align changes with real CRM usage.

Rocket Studio Innovation

Windsor, ON

Full Stack Developer – Intern

Summer 2024

- Developed and enhanced a company wide interview scheduling application, implementing status-based color coding, chronological ordering, and calendar-based lookup to improve interview visibility and scheduling efficiency.
- Implemented code refactoring initiatives to reduce technical debt and optimize performance for long-term scalability.
- Discovered, troubleshooted, and resolved critical front-end and back-end issues, improving application responsiveness and reliability.
- Collaborated in an Agile development environment, delivering features while adhering to engineering best practices.

Extracurricular Involvement

Tutor, United Way

- Delivered over 200 hours of one-on-one tutoring, tailoring sessions to the needs and learning habits of the students.

Projects

Multi-Agent Traffic Simulation *Python, RLLib, Gymnasium, NumPy, Multi-Agent RL*

- Building a city-scale traffic simulation where autonomous car agents learn efficient driving policies via reinforcement learning.
- Training cooperative car and traffic-light agents to reduce congestion and optimize intersection timing.
- Designing reward structures and RLLib/Gym-compatible pipelines for scalable training, logging, and tuning while balancing safety, speed, and collision avoidance.

Stock Price Projector (In Progress) *Python, Pandas, NumPy, yfinance*

- Generates buy/sell recommendations using rule-based signals and thresholding; logs trades and outcomes for analysis.
- Implemented walk-forward validation with metrics including cumulative return, Sharpe ratio, and max drawdown.
- Integrated financial APIs for historical/real-time data; reproducible research via Jupyter notebooks.
- *Roadmap:* add ML models (Gradient Boosting/LSTM), hyperparameter tuning, risk controls, and paper-trading execution.

Personal Website *React, Three.js, React-Three-Fiber, JavaScript, HTML/CSS, Vercel*

- Built a responsive 3D portfolio using React-Three-Fiber and Three.js to showcase projects and technical skills.
- Deployed on Vercel with optimized performance, animations, and mobile support.

Full Stack Password Manager *React, Python, JavaScript, SQL, HTML/CSS*

- Built a secure password manager with encrypted credential storage and hashed authentication.
- Developed a React front end and Python/SQL back end for efficient, user-friendly password management.

Technical Skills

Languages: TypeScript, Python, JavaScript, Java, C, SQL, HTML/CSS

Frameworks and Libraries: React, Angular, Three.js, NEAT, Next.js

AI/ML: NumPy, scikit-learn, PyTorch, RLLib

Tools: Git, Docker, Visual Studio Code,

Methodologies: Agile, OOP, MVC, REST APIs, Microservices

AI/ML Concepts: Reinforcement Learning, Multi-Agent Reinforcement Learning (MARL)